MODULE 27 · DOMAIN 4 - IMPLEMENT TEXT ANALYSIS SOLUTIONS

Translation

Azure Translator's NMT and LLM options, tone and gender, adaptive custom translation, Custom Translator, document translation, and LLM translation flows.

DOMAIN 4 - Text analysis WEIGHT 10-15% MODULE 27 of 39

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Translation is a dial from speed to nuance: NMT for volume, LLMs for tone and context, adaptation for your terminology.

Five ways to translate on Azure

OPTION	WHAT IT IS	STATUS
Text translation	Translate strings with NMT or an LLM	API 2026-06-06, GA
Document translation	Translate whole files, keeping layout	API 2026-03-01, GA
Custom Translator	Train a dedicated NMT model on your data	GA
Adaptive custom translation	Guide LLM output with reference pairs	Playground GA, API preview
Your own LLM flow	A model and prompt in your application	Build it yourself

ON THE EXAM

The first four are Azure Translator in Foundry Tools. The fifth is the study guide's 'LLM-powered translation flow'.

NMT or an LLM, per request

NMT (DEFAULT)

Fast and consistent

- The standard Translator engine
- Billed per source character
- Suits real-time and high-volume text
- No Foundry resource needed

LLM DEPLOYMENT

Nuanced and controllable

- Name a deployment, such as GPT-5.1
- Billed per input and output token
- Tone, gender and adaptive translation
- Requires a Foundry resource

ON THE EXAM

LLM data is processed where its deployment type allows: global, data zone or regional, as module 04 showed.

What changed from v3.0

- **Breaking changes** from v3.0. Microsoft says to test thoroughly before migrating.
- Requests use an **inputs** array. Responses return a **value** array.
- Each input lists its **targets**: language, and optionally a deployment, tone and gender.
- Responses report usage, such as **source characters**, or **tokens** for LLM translations.
- Authenticate with a **resource key** or **Microsoft Entra ID**.

> WATCH FOR THIS
>
> Changing only the api-version on v3.0 code breaks request building and response parsing.

One sentence, two formal Spanish variants

```
{"inputs": [{
  "text": "Doctor is available next Monday. Do you want to schedule an appointment?",
  "language": "en",
  "targets": [
    {"language": "es", "deploymentName": "gpt-translate",
     "tone": "formal", "gender": "female"},
    {"language": "es", "deploymentName": "gpt-translate",
     "tone": "formal", "gender": "male"}]}]}

# value[0].translations: "La doctora esta disponible..." and "El doctor esta disponible..."
```

ON THE EXAM

English 'doctor' has no gender. Spanish needs one, so the gender control decides 'la doctora' or 'el doctor'.

Tone and gender

CONTROL	VALUES	SOLVES
`tone`	formal, informal, neutral	Politeness levels, like 'usted' or 'tu' in Spanish
`gender`	male, female, neutral	Gendered agreement for words that are neutral in English
`deploymentName`	Your LLM deployment	Selects an LLM instead of NMT

ON THE EXAM

Microsoft asks that tone and gender controls be used in line with responsible AI practices.

Tone and gender controls need an **LLM deployment**. A request without one gets **NMT**, which can't honour them.

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Adaptive custom translation or Custom Translator?

ADAPTIVE CUSTOM TRANSLATION

Guide an LLM

- As few as 5 sentence pairs, up to 10,000
- Ready in minutes, no training
- Retrieves similar pairs per request
- Fast-changing or low-volume content

CUSTOM TRANSLATOR

Train an NMT model

- Typically 10,000 or more sentence pairs
- Training can take up to 48 hours
- Retrain and redeploy to update
- High-volume, strict terminology

THE TIEBREAKER

Microsoft's examples: support tickets suit adaptive translation; legal contracts suit Custom Translator.

Building an adaptive dataset

- Upload **5 to 10,000** aligned source and target pairs, in **TMX or TSV** files.
- Each segment must be **250 characters or fewer**. Longer segments are rejected.
- The dataset is ready **in minutes**; reference it by its index ID in translation requests.
- Or pass up to **five reference translations** directly in a request.
- Requires a **Foundry resource**. The playground is GA; the dataset API is in preview.

> ON THE EXAM
>
> Upload failing on some files? Check for segments over 250 characters, using the import job status.

Synchronous or asynchronous?

SYNCHRONOUS

One file, in memory

- Send one document, get it back translated
- No Blob Storage involved
- Suits confidential single files
- The only mode in the Foundry portal

ASYNCHRONOUS BATCH

Many or large files

- Source and target Blob containers
- Storage access by SAS token or managed identity
- Submit a job, poll status, download
- Suits libraries and large files

ON THE EXAM

Both keep the original layout and format, and Microsoft prices them at the same rate.

Document translation features

- Keeps **layout and formatting**; supports **glossaries** and **custom models**.
- **Automatically detects** the source language of each document.
- New: **image translation** for JPEG, PNG, BMP and WebP files, with text rendered back in.
- New: **PDFs through Document Intelligence**, for better layout. Batch only.
- New: text in **images inside Word and PowerPoint** files. Batch only.

> ON THE EXAM
>
> A glossary fixes specific terms in document translation, like product names that must never change.

Translating with your own model and prompt

- Suits **combined tasks**: translate and summarise, translate and extract, or translate inside an agent.
- Put a **glossary** and a **do-not-translate list**, such as brand names, in the instructions.
- Tell the model to **preserve placeholders and markup**, like {name} or HTML tags.
- Return **structured output**, such as source and target side by side, for review.
- Evaluate with **BLEU and METEOR** plus **fluency**, as module 15 showed.

> ON THE EXAM
>
> If the only job is translation, Translator's LLM option usually beats building and maintaining your own prompt.

Requirement to approach

REQUIREMENT	APPROACH
Millions of product reviews, lowest cost	Text translation, NMT
Formal tone and correct gender agreement	Text translation with an LLM deployment
Terminology that changes weekly	Adaptive custom translation
Legal contracts, strict terms, high volume	Custom Translator
One confidential contract, no storage	Synchronous document translation
A library of manuals, layout preserved	Asynchronous batch document translation
Translate and summarise in one step	Your own LLM flow

ON THE EXAM

Place each requirement on the dial: volume, nuance, terminology, file, or combined task.

> A clinic sends appointment reminders in Spanish. Patients must be addressed formally, and job titles like 'doctor' must match the practitioner's gender.
>
> Its current Translator requests use the default engine, and reminders often sound informal or use the wrong gender.

What should the team change?

A. Name an LLM deployment in each target, with tone and gender set.

B. Train Custom Translator on 10,000 formal Spanish sentence pairs.

C. Use synchronous document translation for each reminder message.

D. Add 'formal' and 'female' as glossary entries for the NMT engine.

◆ PAUSE AND DECIDE

What should the team change?

A LLM deployment, tone and gender.

Tone formal and gender per practitioner, exactly the controls the reminders need.

B Custom Translator.

Days of training, and still no per-message gender choice.

C Document translation.

For files. The default engine still ignores tone and gender.

D Glossary entries.

Glossaries fix terms, not register or grammatical agreement.

THE SIGNAL

Formal register plus gender agreement per message: LLM translation controls.

A law firm translates individual confidential contracts, as Word files, from German to English, keeping the original formatting.

Its security policy forbids uploading client documents to any shared storage account, even temporarily.

What should the firm use?

A Asynchronous batch translation, with SAS tokens on the containers.

B Text translation, pasting the text of each contract into the request.

C Custom Translator, trained on the firm's past contract translations.

D Synchronous document translation, one Word file per request.

◆ PAUSE AND DECIDE

What should the firm use?

A Async batch with SAS.

Requires Blob Storage containers, which the policy forbids.

B Paste into text translation.

No storage, but the Word formatting is lost.

C Custom Translator.

Improves terminology, but doesn't decide how files are handled.

D Synchronous document translation.

One file in, translated file out, with layout kept and no storage.

THE SIGNAL

Single confidential files, formatting kept, no storage: synchronous document translation.

A software company translates support tickets. Product and feature names change almost every week.

The team has 300 approved English-to-Japanese sentence pairs, and needs new terms reflected on the same day.

What should the team use?

A Custom Translator, retrained each week on the updated sentence pairs.

B An adaptive dataset from the pairs, used with an LLM deployment.

C A monthly supervised fine-tuning job on a GPT model with the pairs.

D NMT text translation, with no customisation, to keep costs low.

◆ PAUSE AND DECIDE

What should the team use?

A. Weekly Custom Translator.
Typically needs 10,000 pairs, and training can take up to 48 hours.

B. Adaptive dataset with an LLM.
Works from a few hundred pairs, updated and applied within minutes.

C. Monthly fine-tuning.
Monthly misses 'same day', and it's heavier than the problem needs.

D. Plain NMT.
Cheap, but ignores the company's product names entirely.

THE SIGNAL

Few pairs, weekly change, same-day updates: adaptive custom translation, Microsoft's own support-ticket example.

> A team moves an app from Translator v3.0 to API version 2026-06-06 by changing only the api-version value in its requests.
>
> Every request now fails validation, and the response-parsing code also breaks on the few that return.

What should the team change?

A Add the Ocp-Apim-Subscription-Region header to every request.

B Switch authentication from the resource key to Microsoft Entra ID.

C Build requests with inputs and targets; read results from value.

D Move the app to the 2025-10-01-preview version of the API instead.

◆ PAUSE AND DECIDE

What should the team change?

A Region header.

An authentication detail. It doesn't change the request and response schema.

B Entra ID.

Both key and Entra ID work on 2026-06-06. Auth isn't the failure.

C New request and response schema.

2026-06-06 uses inputs, targets and value: breaking changes from v3.0.

D The preview version.

Moves away from the GA version, and previews aren't for production.

THE SIGNAL

Both requests and parsing broke after a version change: the schema changed, not the credentials.

Translation in five lines

- **NMT** for fast, cheap volume; an **LLM deployment** for tone, gender and nuance.
- **API 2026-06-06** breaks v3.0: inputs, targets, value.
- **Adaptive translation:** minutes, few pairs. **Custom Translator:** 10,000 pairs, up to 48 hours.
- **Documents:** synchronous for one file without storage, batch for many or large files.
- **Your own LLM flow** for combined tasks, with glossaries, placeholders and evaluation.

> ON THE EXAM
>
> Next, module 28: speech for agents, including speech to text, text to speech and speech translation.